6 October 1981

MILITARY SPECIFICATION

IMINE CURING/BONDING AGENT

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification establishes the requirements for two types of imine curing/bonding agents to be used as ingredients in case liners and propellants of solid rocket motors.

1.2 Classification. The imine curing/bonding agents shall be of the following types:

Type I:	1, 1'-(1,3-phenylene dicarbonyl) bis 2-methyl-aziridine
Type II:	1, 1',1"-(1,3,5-benzenetriyltricarbonyl) tris 2-ethyl-aziridine

2. APPLICABLE DOCUMENTS.

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specified procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD) Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6810

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM)

ASTM E 203	Water Using Karl Fischer Reagent, Test For.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS.

3.1 Product characteristics. The imine curing/bonding agents shall be amber, viscous liquids.

3.2 Chemical properties. Chemical properties of the imine curing/bonding agents shall conform to Table I.

TABLE I. Chemical properties.

Property	Type I		Type II	
	Min	Max	Min	Max
Imine equivalent weight, g/g equivalent	129	145	135	155
Hydrolyzable chlorine, percent	---	0.9	---	0.9
Moisture, percent	---	0.25	---	0.25

3.3 Environmental conditions. The imine curing/bonding agents shall be sealed in moisture-proof containers at all times during storage. Storage and shipment shall be at a temperature of -18 degrees Celsius (°C) or below.

3.4 Stability. When stored in accordance with 3.3, the imine curing/bonding agents shall have sufficient chemical stability to permit storage for a minimum of 12 months without exceeding 150 grams/gram (g/g) equivalent for Type I and without exceeding 170 g/g equivalent for Type II when retested in accordance with 4.3.1.

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3.5 Toxic products and safety. Safety regulations and guidelines applicable to the use of imine curing/bonding agents should be complied with to preclude personal injury and damage to equipment and facilities.

3.6 Workmanship. Workmanship shall be such that the imine curing/bonding agents are uniform and of consistent high quality.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2.1), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to perscribed requirements.

4.2 Classification of tests. Inspection and testing of the imine curing/bonding agents shall be classified as follows.

4.2.1 Quality conformance inspections and tests. Quality conformance inspections and tests shall consist of the following:

a. Tests of Table I properties (see 4.3).

b. Inspection of filled containers (see 4.5.1).

c. Visual inspection (see 4.5.2).

4.2.1.1 Sampling. A representative sample large enough to perform all tests and a retain sample shall be taken from each lot of material.

4.2.1.2 Acceptance criteria. Failure of any sample to pass all of the requirements specified herein shall cause rejection of the lot represented.

4.3 Test methods. Tests shall be performed using apparatus, reagents, and procedures specified herein. The use of alternate apparatus, reagents, or procedures shall require prior written approval of the procuring activity.

4.3.1 Imine equivalent weight.

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4.3.1.1 Determination of imine equivalent weight. Add 50 milliliters (ml) of methanolic potassium thiocyanate (KSCN) solution (prepared by dissolving 160 grams (gm) of potassium thiocyanate in methanol and diluting to 1 liter) into a 125-ml Erlenmeyer flask fitted with a nitrogen purge. Pipet 10.00 ml of 0.4 Normal (N) methanolic hydrochloric acid (prepared by thoroughly mixing 34 ml of concentrated hydrochloric acid in methanol and diluting to 1 liter) into the flask and thoroughly mix the two solutions. Weigh, to the nearest 0.1 milligram (mg), 0.3 to 0.4 gm of sample into the flask and stir 5 to 6 minutes. Add 3 drops each of 0.1 percent alcoholic bromothymol blue sodium salt and 0.1 percent alcoholic phenol red sodium salt indicators. Titrate to a purple end point with standardized 0.1 N alcoholic potassium hydroxide (KOH) (prepared by dissolving 6 gm of potassium hydroxide in methanol and diluting to 1 liter). Determine two reagent blanks following the above procedure.

Calculation:

$$\text{Imine equivalent weight, g/g equivalent} = \frac{W \times 1000}{(V_1 - V_2)\,N}$$

Where: V_1 = volume of alcoholic KOH used for blank, ml

V_2 = volume of alcoholic KOH used for sample, ml

N = normality of alcoholic KOH

W = weight of sample, gm

Report the imine equivalent weight to the nearest whole unit.

4.3.2 Hydrolyzable chlorine.

4.3.2.1 Determination of hydrolyzable chlorine. Weigh, to the nearest 0.1 milligram (mg), a 1 gm sample into a 250-ml flask with a ground glass neck. Add 50 ml of 0.5 N alcoholic potassium hydroxide (prepared by dissolving 30 gm of potassium hydroxide in methanol and diluting to 1 liter) and a few boiling chips to the flask. Insert a condenser into the flask and heat to maintain a steady drop rate for 45 minutes (start timing when first reflux begins). Allow solution to cool, rinse the condenser with approximately 50 ml of methanol and disconnect flask. Quantitatively transfer the solution to a 400-ml beaker using small portions of methanol. Using methyl red indicator, neutralize the solution by dropwise addition of concentrated nitric acid solution, then add 3 ml excess. Titrate potentiometrically with standardized 0.02 N silver nitrate (prepared by dissolving 3.4 gm of silver nitrate in distilled water and diluting to 1 liter) using a combination of silver-billet/double junction glass electrode and automatic titrator. Run a reagent blank using the same method.

Calculation:

$$\text{Hydrolyzable chlorine, percent} = \frac{3.545(V_1 - V_2)\,N}{W}$$

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Where: V_1 = volume of standardized 0.02 N silver nitrate used for sample, ml

V_2 = volume of standardized 0.02 N silver nitrate used for blanks, ml

N = normality of standardized 0.02 N silver nitrate

W = weight of sample, gm

3.545 = 0.1 atomic weight of chlorine

Report hydrolyzable chlorine to the nearest 0.1 percent

4.3.3 Moisture. The moisture content shall be determined in accordance with ASTM E 203 using a 5.0 plus or minus 0.1 gm (weighed to the nearest 0.1 mg) sample titrated to a 30-second end point. Solvent shall be methanol-chloroform, 1:1 by volume. Report moisture content to the nearest 0.01 percent.

4.4 Stability. During the first 12 months of storage in accordance with 3.3, the material shall conform to Table I. Shelf life may be extended for an additional 6-month period by verifying conformance to Table I.

4.5 Examinations.

4.5.1 Inspection of filled containers. All filled containers shall be inspected prior to shipment or use for accuracy of markings and for defects in containers and closures. All defective containers and closures shall be repaired or replaced, and contents therein shall be reinspected prior to shipment or use.

4.5.2 Visual inspection. All samples shall be visually inspected to determine conformance to the requirements of 3.6.

4.6 Records. Certification and test data shall be prepared as required by the procuring activity (see 6.2.2).

5. PACKAGING.

5.1 Preservation-packaging and packing. Unless otherwise specified in the contract or purchase order (see 6.2.1), packaging and packing of the imine curing/bonding agent shall be in accordance with commercial practice to insure carrier acceptance and shall be of such construction and materials that the contents will be adequately protected against loss or contamination. Protection from loss shall include packing in dry ice to prevent temperature degradation during transit.

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5.2 Marking for shipment. Unless otherwise specified in the contract or purchase order (see 6.2.1), each shipping container shall be marked in accordance with the requirements of MIL-STD-129. Container markings shall include the following:

a. The supplier's lot number.

b. Storage conditions.

CAUTION

SHIP AND STORE AT TEMPERATURES BELOW -18°C

c. Toxicity warning.

d. Date of manufacture.

e. Manufacturers Code Ident.

f. Net weight of contents.

g. Material name.

6. NOTES AND CONCLUDING MATERIAL.

6.1 Intended use. The intended use of the materials described herein are as a bonding agent between ammonium perchlorate and polymer for Type I, and as a bond promoter in a liner composition for Type II.

6.2 Ordering data.

6.2.1 Procurement requirements. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Responsibility for inspection and inspection facilities, if different than 4.1.

c. Special packaging, packing, or marking requirements, if applicable (see Section 5).

6.2.2 Data requirements. When this specification is used in a procurement which incorporates a Contract Data Requirements List (DD Form 1423) and invokes the provisions of 7-104.9(n) of the Defense Acquisition Regulations (DAR), the data requirements identified below will be developed as specified by an approved Data Item Description (DID) (DD Form 1664) and delivered in accordance with the approved DD Form 1423 incorporated into the contract. When the provisions of DAR-7-104.9(n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraphs:

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Paragraph	Data Requirement	Applicable DID
4.6	Certification Test Data	UDI-A-23264B DI-T-4024

(Copies of DIDs required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

6.3 Suggested source of supply. Products that have met the requirements of this specification in past procurement actions are manufactured by Minnesota Mining and Manufacturing, Code Ident 28112 as HX 752 (Type I) and HX 868 (Type II). This information is for the convenience of the procuring activity and is not to be construed as a waiver of any requirement of this specification nor as any limitation of additional potential source of supply.

Preparing activity:
Navy - AS

(Project 6810-NB17)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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